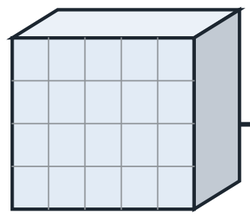
Ray Tracing Attention

Optical attention, recurrent fields and signed MLP transmission

(a) Ray Tracing Attention



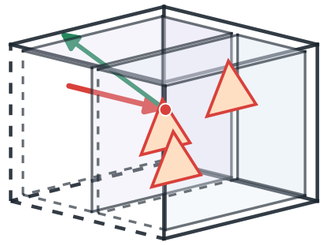
$X \in \mathbb{R}^{T \times D}$

Projection + RoPE

Q / K / gate / geometry

Complete blocks P_b

Causal BVH query



$s_{tj} = q_t^{\mathsf{T}} k_j / \sqrt{d}, \quad j \le t$

Light response

$I_{tj} = e^{c_t + s_{tj}}$

$g_j = \mathcal{D}(\gamma_j)$

$$Z_t = \frac{\sum_{j \le t} e^{s_{tj}} g_j}{\sum_{j \le t} e^{s_{tj}}}$$

Dual detector

$S_t = \sum_j I_{tj} g_j$

$R_t = \sum_j I_{tj}$

Restore + project

$Z_t = S_t / R_t$

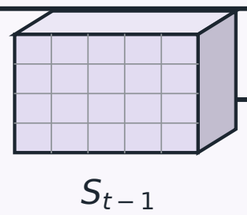
$O_t = W_O(\text{gate} \odot Z_t)$

$P^{-1}(\oplus_b P_b O_t)$

(b) Recurrent light-field operator

Current optical input

$k_t,\ g_t,\ \alpha_t,\ \beta_t$

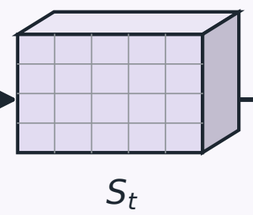


S_{t-1}

Decay + correction + write

$S_t^- = \alpha_t S_{t-1}, \quad \Delta_t = \beta_t \odot (g_t - (S_t^-)^{\mathsf{T}} k_t)$

$S_t = S_t^- + k_t \Delta_t^{\mathsf{T}}$



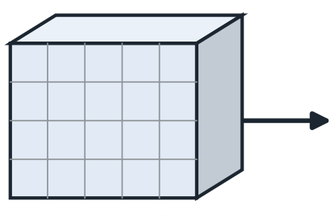
S_t

Query + detection

$o_t = S_t^{\mathsf{T}} q_t / \sqrt{d_k}$

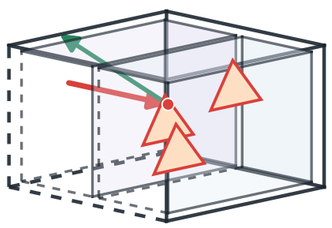
q_t

(c) Ray Tracing MLP



$h \in \mathbb{R}^D$

Feature geometry

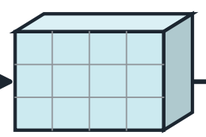


$W_g,\ W_u$

$g = \mathcal{O}_{W_g}(h)$

SiLU

$u = \mathcal{O}_{W_u}(h)$

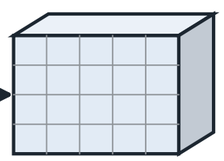


$z = g_{\text{SiLU}} \odot u$

Down response

$y = \mathcal{O}_{W_d}(z)$

W_d



$P^{-1}(\oplus_b P_b y)$

Signed detection: $E^+ - E^- = \sum_c x_c w_{rc}$